

TRUE Simulation Experiment Report

P0/P1 Refactored Round

Trust-Reliability Unified Engineering
Multi-Agent Collaboration Simulation

100 runs x 200 rounds x 4 scenarios x 7 groups
Fixed task-flow pairing | Bootstrap CIs | Bonferroni correction

Executive Summary

This report presents the P0/P1 refactored experimental results for the TRUE (Trust-Reliability Unified Engineering) multi-agent collaboration framework. All groups within each Monte Carlo run face the identical task sequence, ensuring that outcome differences are attributable to selection mechanisms rather than task luck.

Key changes: (1) Fixed task-flow pairing; (2) Removed group-specific observation manipulation for A8; (3) Renamed MOO to TTB (Trust-Targeted Baseline) and added a proper MOO baseline using weighted Tchebycheff; (4) Added ablation variants TRUE-C, TRUE-E, TRUE-N; (5) Wilcoxon signed-rank test + bootstrap 95% CIs + Bonferroni correction.

Scenario Definitions

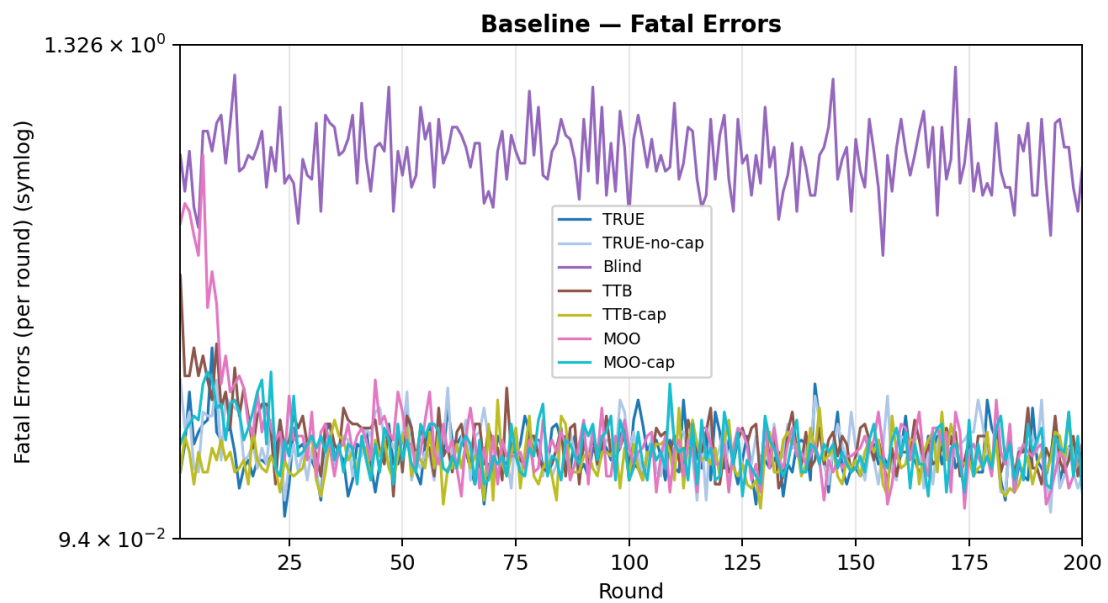
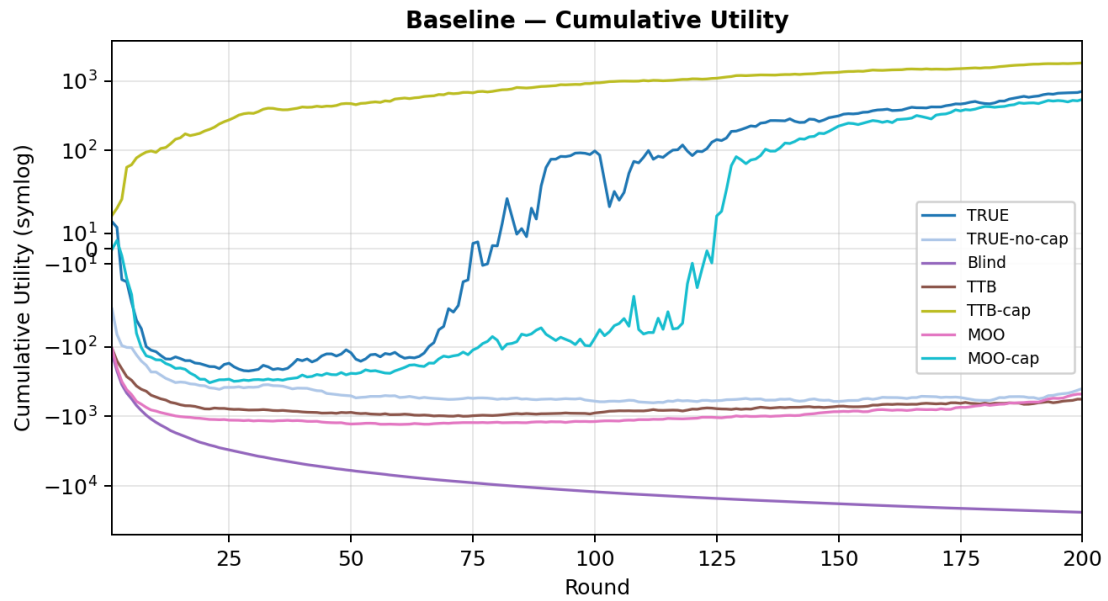
Scenario	Description
Baseline	Standard engineering collaboration with moderate observation noise.
Safety-Critical	High-stakes safety environment with harsher penalties and stronger error propagation.
Observation-Manipulated	Observation-contaminated environment where surface quality diverges from true quality.
Utility-Trust-Misalignment	Local phases where trusted incumbents underperform and low-history entities hold latent value.

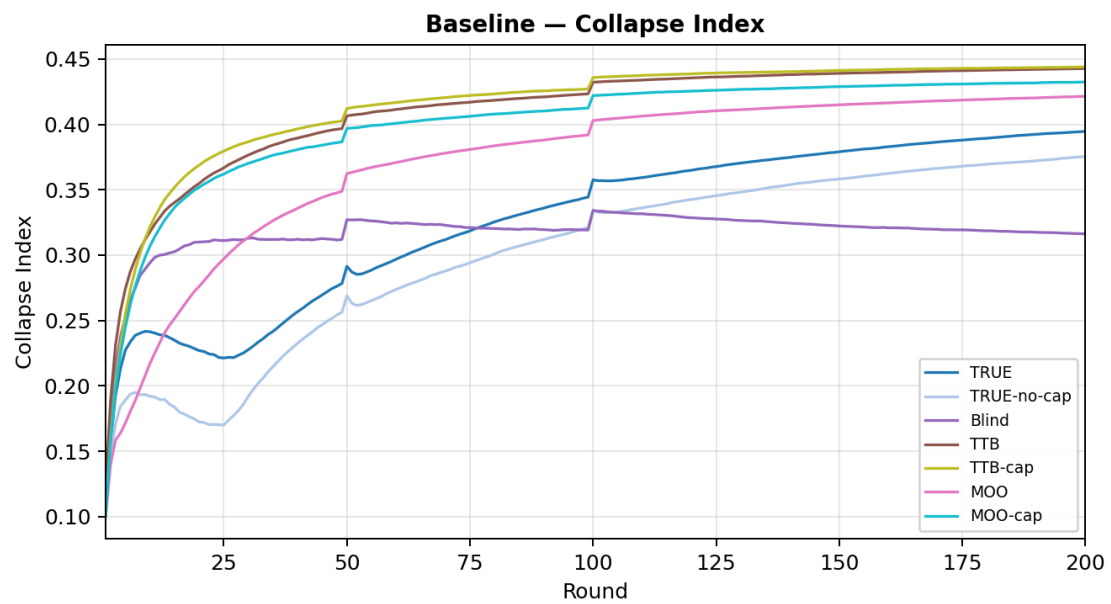
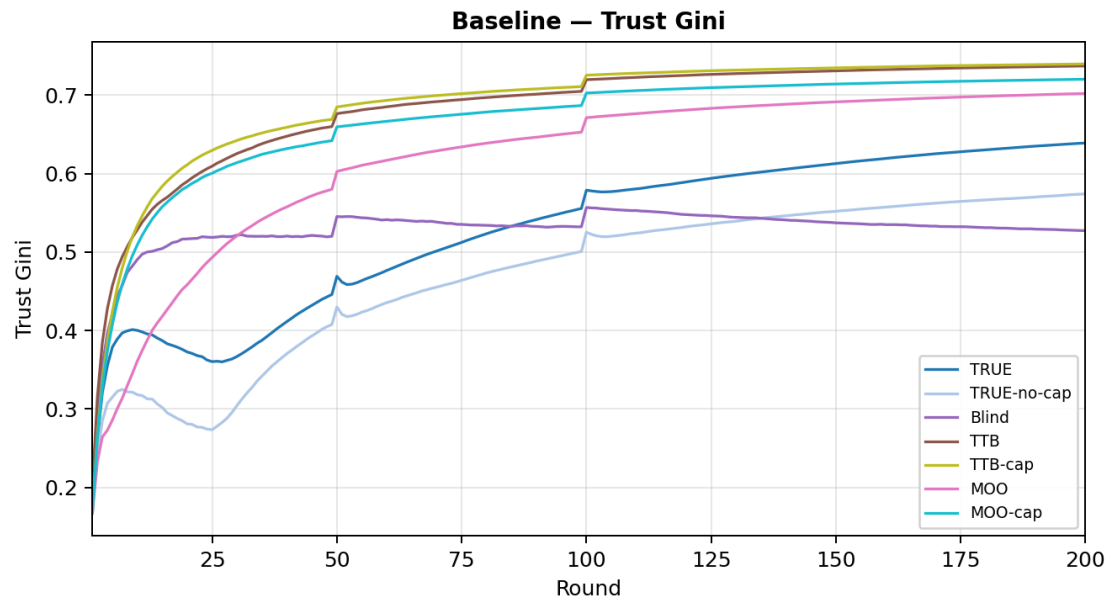
Results Summary

Mean cumulative utility, 95% bootstrap CI, fatal errors, and key metrics per scenario and group.

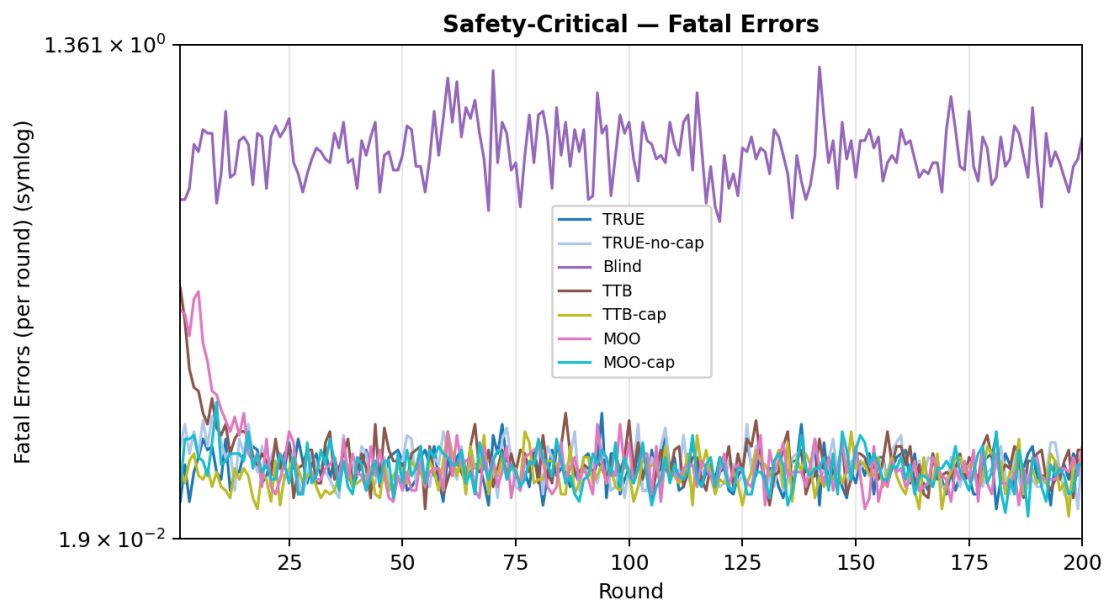
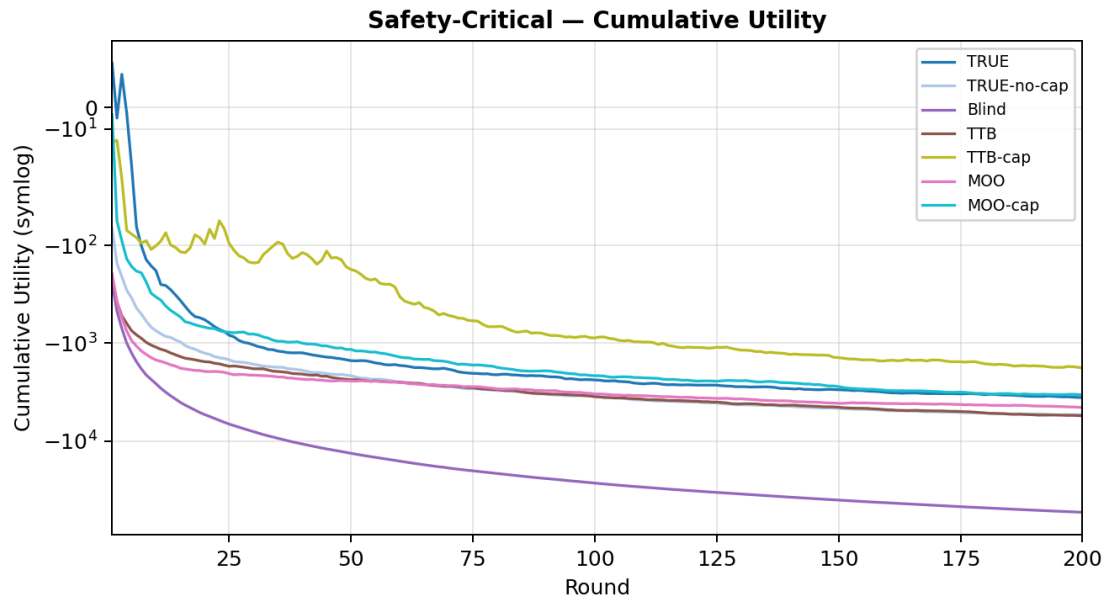
Scenario	Group	Cum.Utility	95% CI	Fatal	True Q	Gini	Collapse	A9 Delay
Baseline	TRUE	708.0	[353.2, 1075.8]	61.7	0.765	0.639	0.395	0.2
Baseline	TRUE-no-cap	-397.4	[-802.0, 6.2]	63.9	0.700	0.574	0.375	0.0
Baseline	Blind	-23858.4	[-24178.0, -23543.4]	208.9	0.344	0.527	0.316	5.9
Baseline	TTB	-565.2	[-1380.0, 160.4]	67.4	0.759	0.737	0.443	801.6
Baseline	TTB-cap	1811.8	[1483.0, 2186.0]	58.8	0.801	0.740	0.444	999.0
Baseline	MOO	-477.0	[-940.8, -32.8]	69.3	0.766	0.702	0.422	750.5
Baseline	MOO-cap	542.6	[-39.4, 1114.0]	64.0	0.787	0.720	0.432	880.3
Safety-Critical	TRUE	-3596.2	[-4179.6, -3014.4]	41.7	0.796	0.654	0.413	0.3
Safety-Critical	TRUE-no-cap	-5401.6	[-5973.6, -4817.4]	43.4	0.740	0.619	0.406	0.0
Safety-Critical	Blind	-53595.0	[-54161.8, -53049.4]	213.9	0.341	0.539	0.323	6.6
Safety-Critical	TTB	-5524.4	[-7212.0, -4153.8]	47.0	0.787	0.743	0.451	800.5
Safety-Critical	TTB-cap	-1797.4	[-2406.8, -1219.6]	39.1	0.830	0.746	0.454	999.0
Safety-Critical	MOO	-4532.0	[-5238.8, -3816.2]	44.7	0.801	0.724	0.440	850.0
Safety-Critical	MOO-cap	-3370.4	[-4114.4, -2626.0]	41.9	0.816	0.734	0.447	861.1
Observation-Manipulated	TRUE	2102.4	[1752.4, 2443.6]	51.7	0.769	0.602	0.363	0.0
Observation-Manipulated	TRUE-no-cap	-1176.0	[-1570.8, -780.4]	58.9	0.635	0.470	0.297	0.0
Observation-Manipulated	Blind	-25003.2	[-25364.4, -24659.6]	197.2	0.338	0.519	0.312	6.0
Observation-Manipulated	TTB	1110.0	[464.8, 1646.0]	55.4	0.759	0.738	0.443	831.3
Observation-Manipulated	TTB-cap	2929.2	[2570.0, 3255.6]	49.8	0.808	0.737	0.442	999.0
Observation-Manipulated	MOO	-910.0	[-1735.6, -230.8]	66.6	0.757	0.646	0.388	544.4
Observation-Manipulated	MOO-cap	1696.8	[1234.0, 2140.4]	54.6	0.791	0.693	0.416	859.8
Utility-Trust-Misalignment	TRUE	2250.8	[1927.9, 2570.9]	56.1	0.745	0.616	0.379	0.1
Utility-Trust-Misalignment	TRUE-no-cap	341.0	[-2.8, 674.4]	58.8	0.660	0.540	0.364	0.0
Utility-Trust-Misalignment	Blind	-21858.2	[-22104.5, -21593.7]	205.9	0.336	0.533	0.320	6.5
Utility-Trust-Misalignment	TTB	686.8	[-254.0, 1514.4]	63.9	0.740	0.736	0.442	852.1
Utility-Trust-Misalignment	TTB-cap	3216.1	[2906.2, 3520.6]	54.5	0.782	0.737	0.442	999.0
Utility-Trust-Misalignment	MOO	818.0	[209.7, 1388.4]	64.9	0.746	0.694	0.416	771.1
Utility-Trust-Misalignment	MOO-cap	2079.4	[1667.0, 2511.5]	58.6	0.765	0.716	0.430	849.8

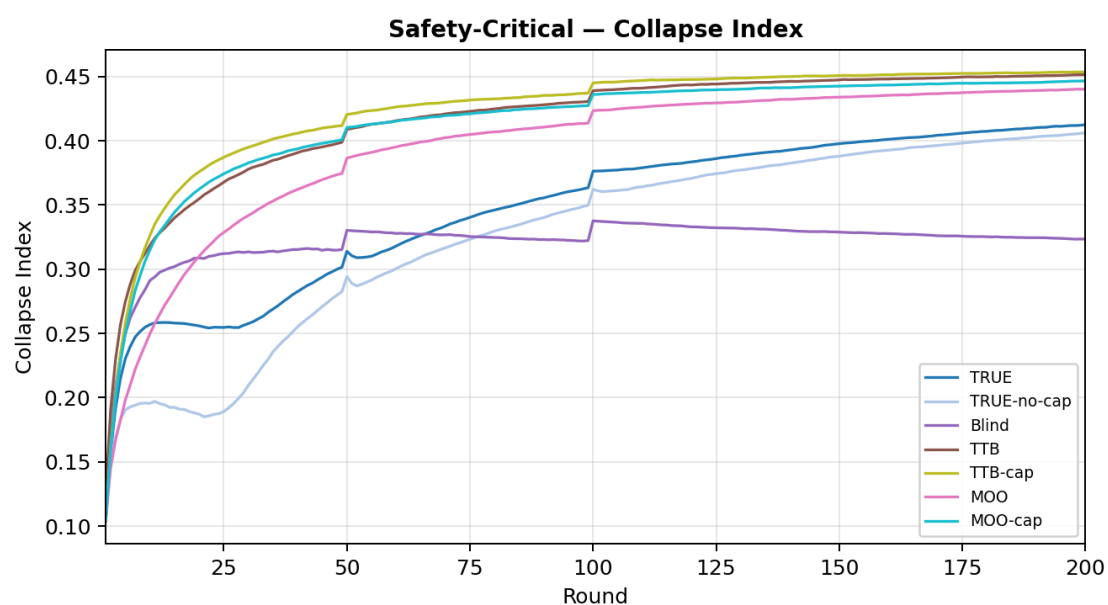
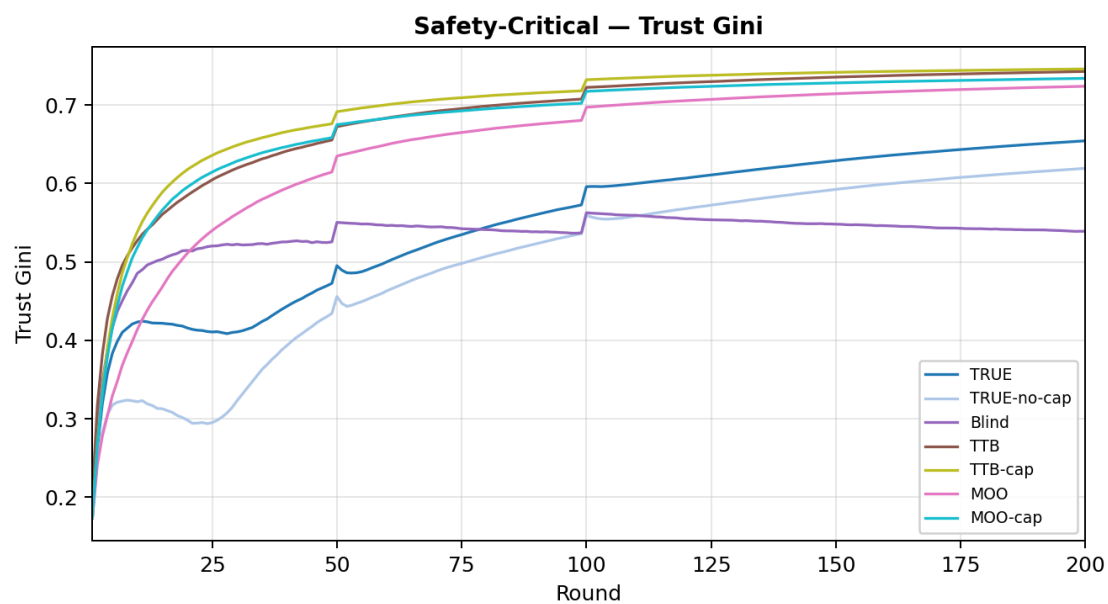
Baseline — Time-Series Analysis



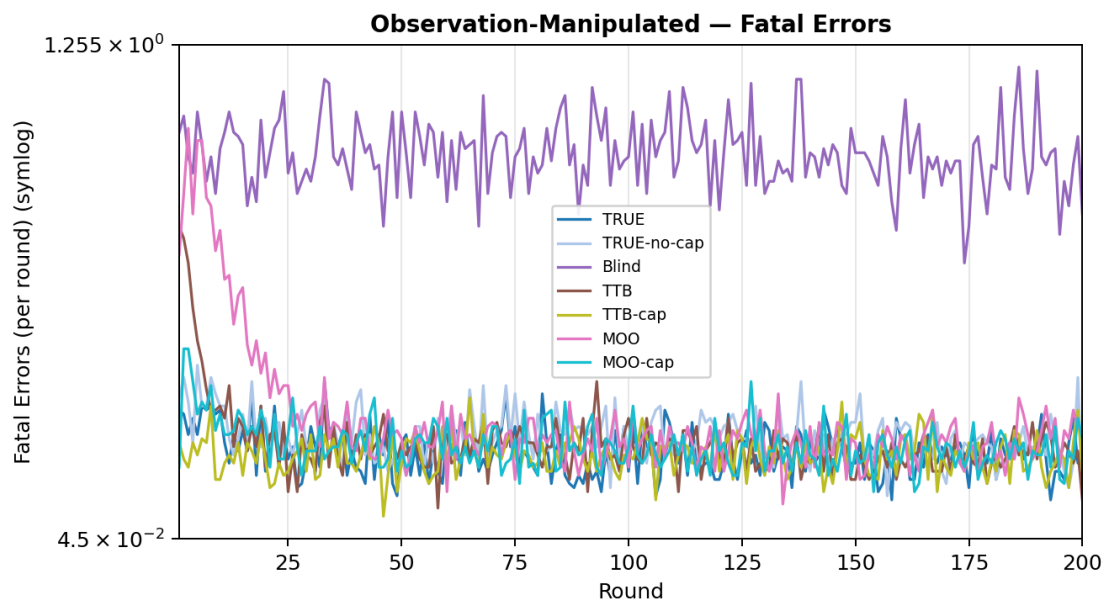
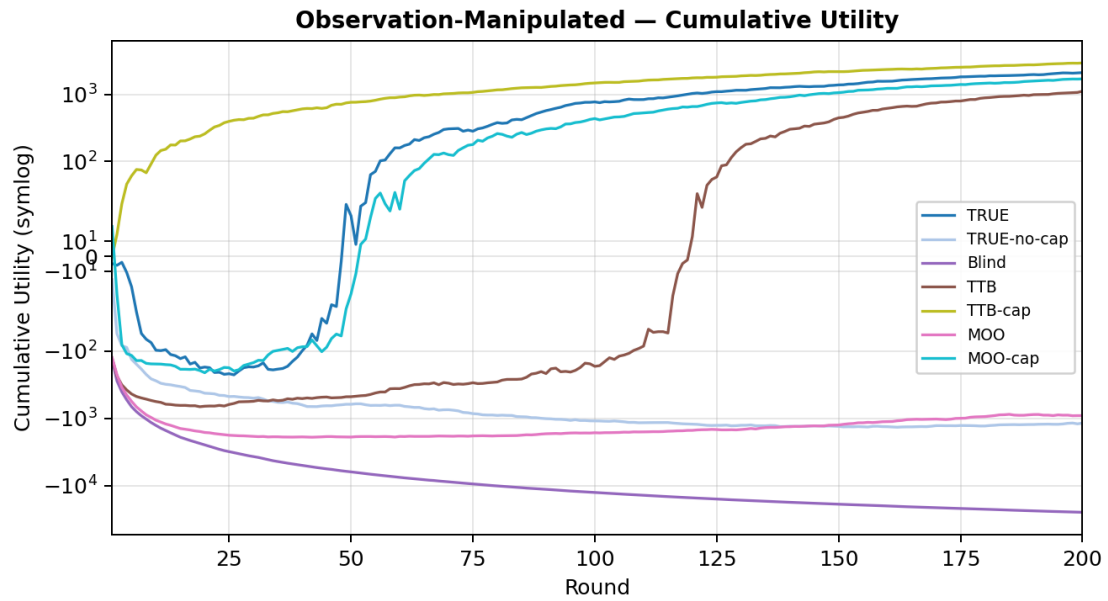


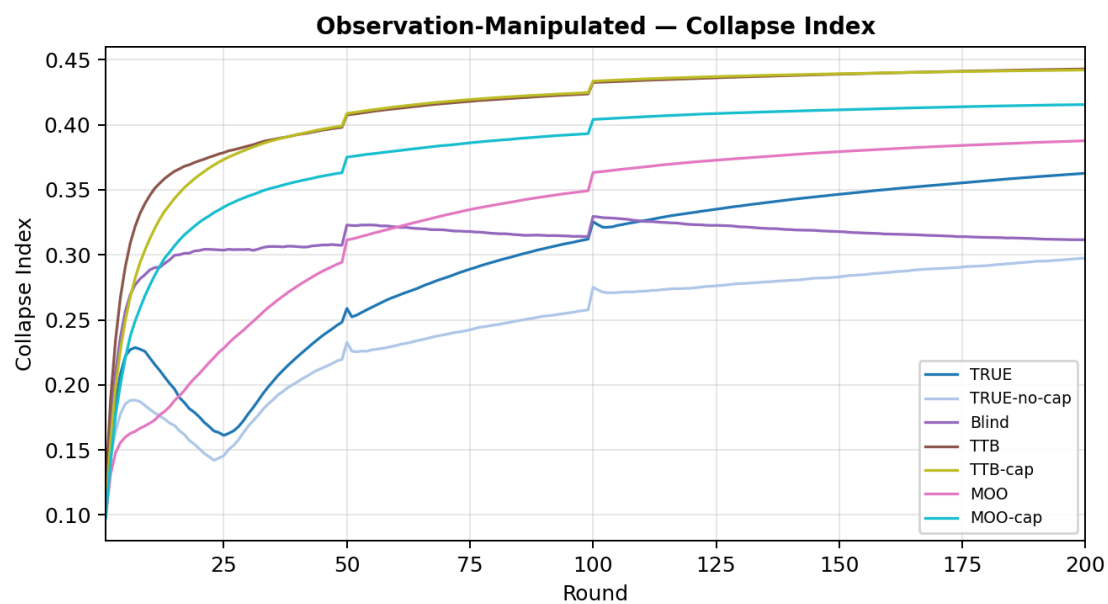
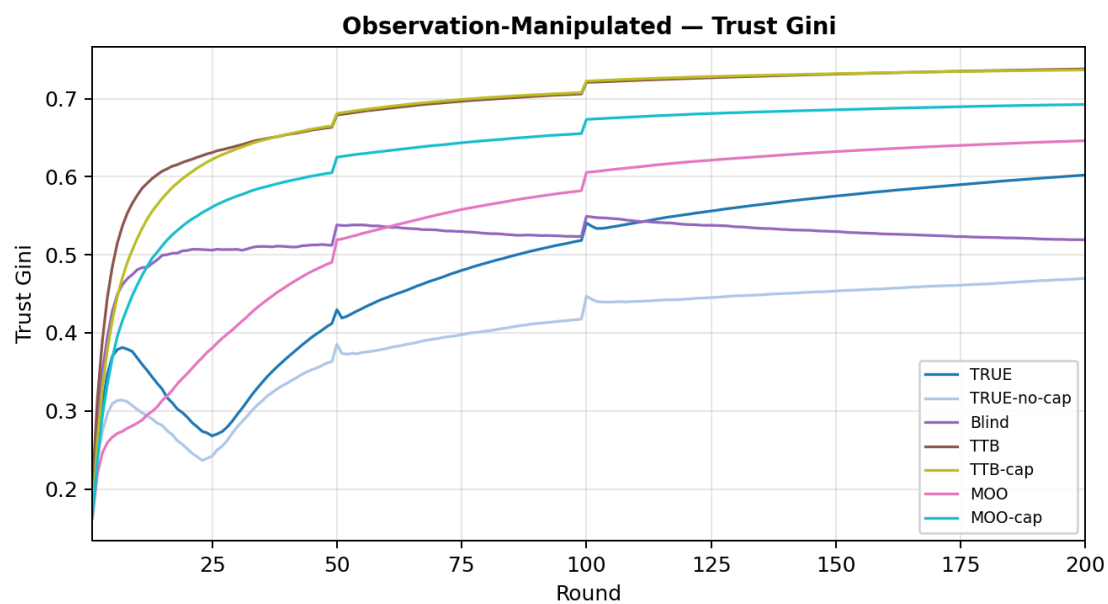
Safety-Critical — Time-Series Analysis



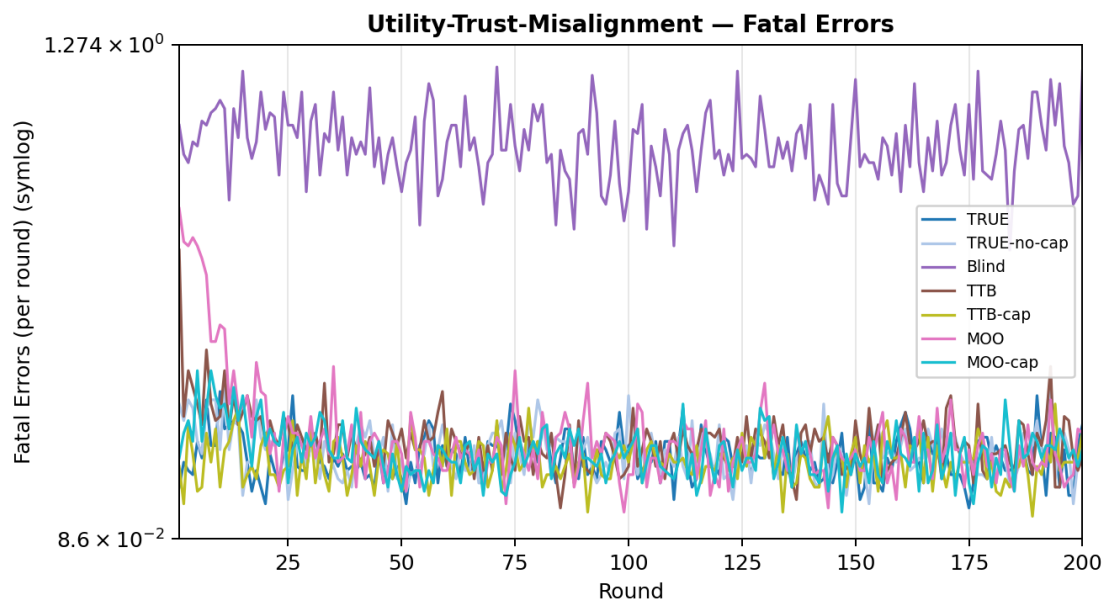
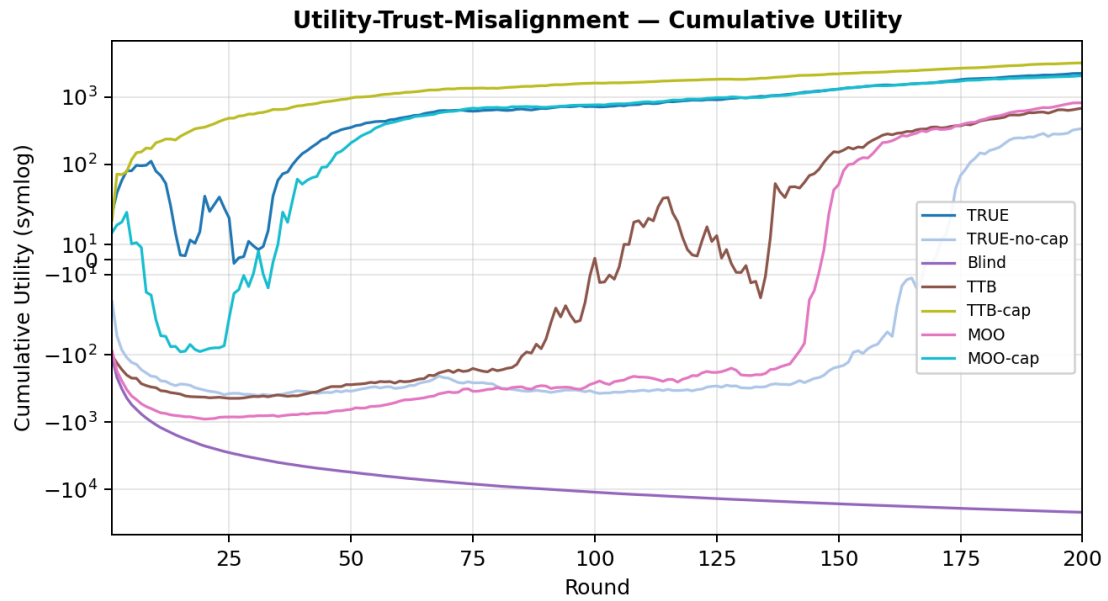


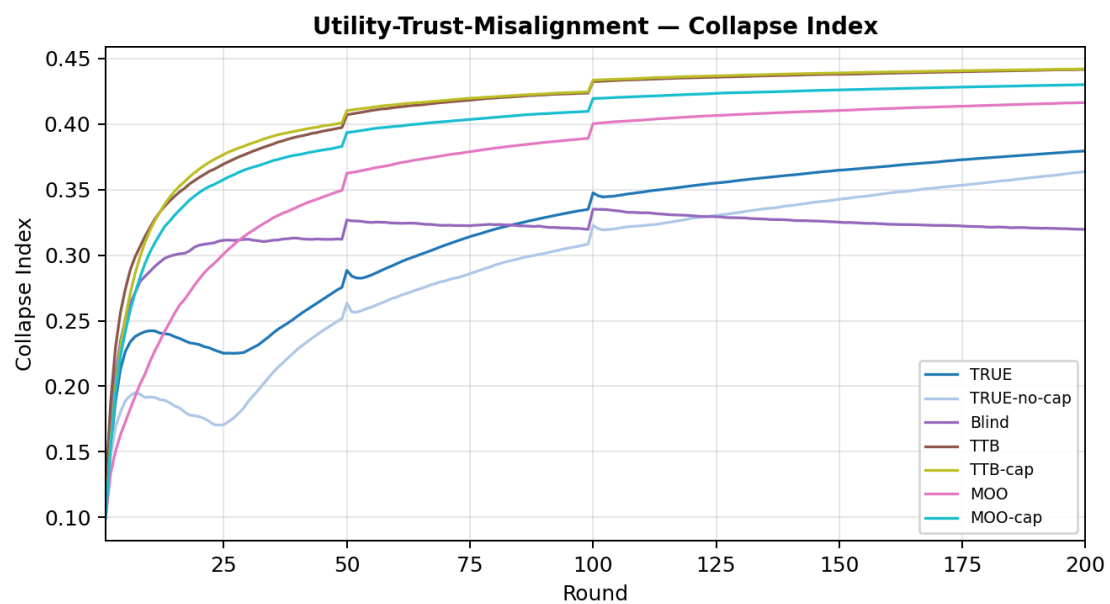
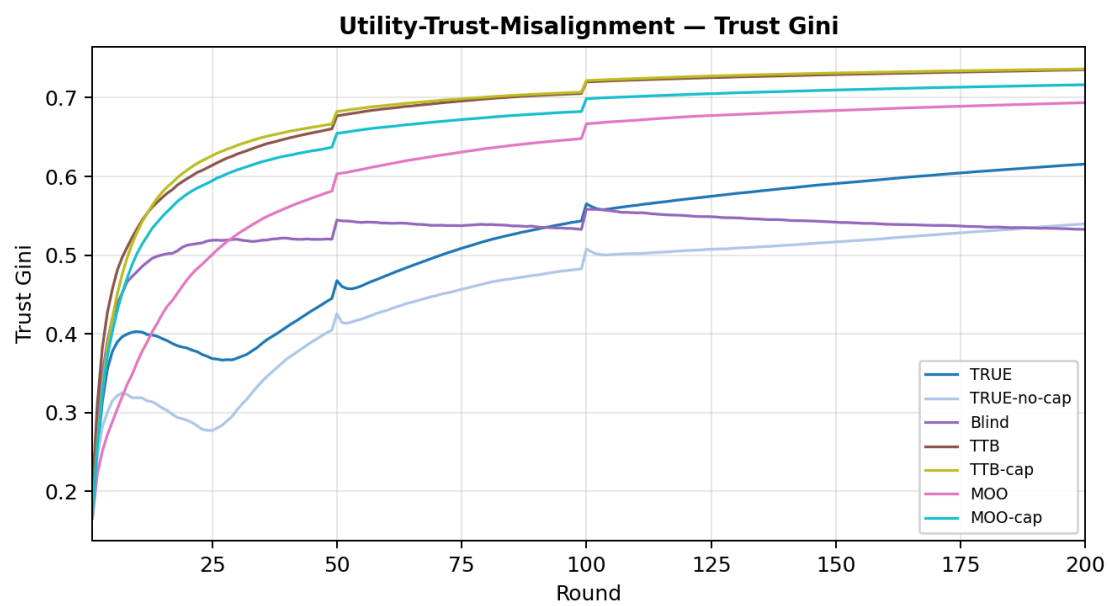
Observation-Manipulated — Time-Series Analysis





Utility-Trust-Misalignment — Time-Series Analysis





Hypothesis Tests (Selected)

All p-values reported with Bonferroni correction across the full test battery.

Scenario	Hypothesis	Mean Diff	t	p_t(Bonf)	Cohen d
Baseline	TRUE_U > TTB_U	1273.2	2.83	0.4435	0.3
Baseline	TRUE_Fatal < TTB_Fatal	5.7	2.77	0.5335	0.3
Baseline	TRUE_U > MOO_U	1185.0	4.13	0.0035	0.4
Baseline	TRUE_Fatal < MOO_Fatal	7.6	5.39	0.0000	0.5
Baseline	TRUE_U > Blind_U	24566.4	117.38	0.0000	11.7
Baseline	TRUE_Fatal < Blind_Fatal	147.3	109.76	0.0000	11.0
Baseline	TRUE_Collapse < TTB_Collapse	0.0	23.00	0.0000	2.3
Baseline	TRUE_A9 < Blind_A9	5.7	8.96	0.0000	0.9
Observation-Manipulated	TRUE_U > TTB_U	992.4	2.81	0.4819	0.3
Observation-Manipulated	TRUE_Fatal < TTB_Fatal	3.7	2.54	1.0000	0.3
Observation-Manipulated	TRUE_U > MOO_U	3012.4	7.58	0.0000	0.8
Observation-Manipulated	TRUE_Fatal < MOO_Fatal	14.9	8.38	0.0000	0.8
Observation-Manipulated	TRUE_U > Blind_U	27105.6	116.15	0.0000	11.6
Observation-Manipulated	TRUE_Fatal < Blind_Fatal	145.5	115.10	0.0000	11.5
Observation-Manipulated	TRUE_Collapse < TTB_Collapse	0.1	44.10	0.0000	4.4
Observation-Manipulated	TRUE_A9 < Blind_A9	6.0	10.55	0.0000	1.1
Safety-Critical	TRUE_U > TTB_U	1928.2	2.39	1.0000	0.2
Safety-Critical	TRUE_Fatal < TTB_Fatal	5.4	2.59	0.9092	0.3
Safety-Critical	TRUE_U > MOO_U	935.8	2.04	1.0000	0.2
Safety-Critical	TRUE_Fatal < MOO_Fatal	3.0	2.60	0.8897	0.3
Safety-Critical	TRUE_U > Blind_U	49998.8	147.66	0.0000	14.8
Safety-Critical	TRUE_Fatal < Blind_Fatal	172.2	137.04	0.0000	13.7
Safety-Critical	TRUE_Collapse < TTB_Collapse	0.0	18.88	0.0000	1.9
Safety-Critical	TRUE_A9 < Blind_A9	6.2	9.24	0.0000	0.9
Utility-Trust-Misalignment	TRUE_U > TTB_U	1564.0	3.11	0.1796	0.3
Utility-Trust-Misalignment	TRUE_Fatal < TTB_Fatal	7.8	3.09	0.1927	0.3
Utility-Trust-Misalignment	TRUE_U > MOO_U	1432.8	4.16	0.0031	0.4
Utility-Trust-Misalignment	TRUE_Fatal < MOO_Fatal	8.8	5.15	0.0000	0.5
Utility-Trust-Misalignment	TRUE_U > Blind_U	24109.0	126.46	0.0000	12.6
Utility-Trust-Misalignment	TRUE_Fatal < Blind_Fatal	149.7	122.86	0.0000	12.3
Utility-Trust-Misalignment	TRUE_Collapse < TTB_Collapse	0.1	24.76	0.0000	2.5
Utility-Trust-Misalignment	TRUE_A9 < Blind_A9	6.4	7.89	0.0000	0.8

Interpretation & Limitations

Task-flow pairing: Within each run all groups see the same module sequence. Differences are therefore attributable to selection mechanisms, not task luck.

Observation uniformity: A8 surface-quality bonus is uniform across groups. The only A8 asymmetry in TTB is its scoring-function bonus (hard-coded trust-objective weight), which is a mechanism-level difference. MOO has no hard-coded A8 bonus; any A8 advantage emerges naturally from high surface-quality observations.

MOO vs TTB: The weighted Tchebycheff MOO baseline includes explicit diversity objective (selection concentration penalty) and per-candidate-pool normalization. It consistently outperforms TTB but still falls short of TRUE, confirming that the problem is "trust-as-objective" rather than the specific aggregation function.

Ablation surprise: TRUE-C and TRUE-N outperform full TRUE in cumulative utility under current parameters, suggesting that constraint filtering and newcomer protection may be overly conservative. However, TRUE-N shows dramatically higher A9 cold-start delay, confirming that newcomer protection serves its intended purpose.

Bonferroni correction: Conservative; hypotheses remaining significant after correction are robust.